# Sparse STVG-R1: evidence and task controls

Finite development diagnostic | 8 exposed queries | frozen STVG-R1-7B | 2026-10-14

# P0-1: same-candidate input representation

Temporal: paired sparse snippets versus two contiguous local boundary windows sampled uniformly at stride four. Spatial: same eight physical frames, paired small panels versus independent 448-square full-scene views. Different evidence/pixel budgets are reported; this is not isolated resolution causality.

| Domain / branch | View | ID same | Order same | Stable | Base correct |
|---|---|---|---|---|---|
| hc2 / T | small | 3/4 | 2/4 | 2/4 | 2/4 |
| hc2 / T | high | 2/4 | 3/4 | 1/4 | 3/4 |
| hc2 / S | small | 2/4 | 3/4 | 1/4 | 3/4 |
| hc2 / S | high | 3/4 | 1/4 | 0/4 | 2/4 |
| vidstg / T | small | 2/4 | 3/4 | 1/4 | 4/4 |
| vidstg / T | high | 1/4 | 4/4 | 1/4 | 4/4 |
| vidstg / S | small | 0/4 | 3/4 | 0/4 | 4/4 |
| vidstg / S | high | 0/4 | 4/4 | 0/4 | 4/4 |

ID/order consistency counts all four fixed pairs per domain. Correctness counts only informative, decisive comparisons; different denominators must not be mixed. Stable Tie/Uncertain is not a usable update direction.

# P0-2: Query-conditioned task controls

Native-style adapts official Target ID / Time range syntax to supplied hypotheses; it is not native open-world benchmark accuracy. Simplified alignment receives partial Query-free subject/action facts plus the same visuals and Query. Boundaries are never judged without Query.

| Domain / branch | Task | ID same | Order same | Stable dir. | Stable correct |
|---|---|---|---|---|---|
| hc2 / T | native | 2/4 | 3/4 | 1/4 | 0/1 |
| hc2 / T | align | 2/4 | 4/4 | 0/4 | 0/0 |
| hc2 / S | native | 3/4 | 1/4 | 0/4 | 0/0 |
| hc2 / S | align | 3/4 | 3/4 | 0/4 | 0/0 |
| vidstg / T | native | 1/4 | 4/4 | 1/4 | 1/1 |
| vidstg / T | align | 0/4 | 2/4 | 0/4 | 0/0 |
| vidstg / S | native | 0/4 | 4/4 | 0/4 | 0/0 |
| vidstg / S | align | 1/4 | 2/4 | 0/4 | 0/0 |

Output validity

| Task | Responses | Accepted | Truncated |
|---|---|---|---|
| native_style | 136 | 136 | 0 |
| facts | 48 | 46 | 0 |
| alignment | 48 | 48 | 0 |

Native time-copy diagnostic: 85/136 answers exactly copied the prompt example's 0.00-to-1.00 range. All 16/16 wrong-query calls kept the base ID preference; replacement queries were byte-distinct but may share visual semantics. Neither statistic alone proves complete Query blindness.

# Coarse geometry and conditional adaptation

| Domain | ID same | Order same | Stable dir. | Stable correct |
|---|---|---|---|---|
| hc2 | 4/4 | 0/4 | 0/4 | 0/0 |
| vidstg | 0/4 | 4/4 | 0/4 | 0/0 |

The coarse control truncates a student box by 55% on its right side. Neither candidate is asserted to be a complete correct referent. Post-seal truth checks quantify actual quality differences. Local candidates are not different object instances.

Conditional update cases: 0. Student fits executed: 0. Zero fallback improvement is not an adaptation measurement. Temporal fitting to Native is a no-op; a reliable Projection target is needed to adapt the original Native logits.

Actual frozen evaluator calls: 232; preprocess + generation 328.11 s; input tokens 786,935; output tokens 6,601; peak allocated GPU 17.43 GiB. Excludes source capture, view decoding/loading, and any student adaptation; not end-to-end TTA cost.

# Limits and decision scope

Only four exposed queries per direction, unknown teacher training overlap, no fresh confirmation or online carry. Stable preferences do not alone establish geometric correctness. Full-scene box-center labels differ from official mask-center IDs; glyph placement uses default PIL anchoring. Physical global offsets are supplied as text for separate clips. All positive, negative, abstaining and unknown outcomes remain in the anonymous JSON export. The finite diagnostic does not settle all MLLM feedback or OPD methods.